

STRUCTURAL EQUATIONS, TREATMENT EFFECTS, AND ECONOMETRIC POLICY EVALUATION

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1. PREPARATION

PROBLEM

- Rubin model: to evaluate the effect of public policies

$$\Delta^{ATE} = \mathbb{E}(Y_1 - Y_0)$$

- Ignorability / Unconfoundedness:

$$(Y(0), Y(1)) \perp W \mid X$$

- Endogeneity Problems

- Self-selection: individuals participate only if they expect sufficiently large returns
- Randomized experiments are often infeasible

- IV (Imbens & Angrist, 1994)

$$Z = 1 \text{ or } 0$$

$$\mathbb{E}(Y_1 - Y_0 \mid \text{compliers})$$

PROBLEM

Two problems:

- Different instrumental variables produce different estimates
 - Compliers are defined according to the specific instrument. (Imbens & Angrist, 1994)
- Policymakers are interested in different (combinations of) treatment effect parameters depending on the question
 - ATE: What is the average gain if everyone receives the treatment?
 - TT: What is the average gain for those who have already received the treatment?
 - LATE: What is the average gain for individuals whose decisions are changed by the instrument?
 - PRTE: How much does social welfare change if policy a' is replaced by policy a ?

$$PRTE = \mathbb{E}(V(Y^a)) - \mathbb{E}(V(Y^{a'}))$$

- Rubin Causal Model

Potential outcomes (Y_0, Y_1) for conditioning variables X :

$$Y_1 = \mu_1(X) + U_1 \quad (\text{treated})$$

$$Y_0 = \mu_0(X) + U_0 \quad (\text{not treated})$$

Receipt of treatment: $D = 1$ (or $D = 0$)

Thus, the observed outcome Y is given by

$$Y = DY_1 + (1 - D)Y_0.$$

- X is observable but U_0, U_1 are not.
- Permit that

$$\text{Cov}(U_g, X) \neq 0, \quad g \in \{0, 1\}.$$

- Cost of receiving treatment

$$C = \mu_C(Z) + U_C$$

- Threshold for participation: intuition, distance, ...
- Z in Imbens and Angrist (1994) can also be interpreted.
- Z can be continuous or discrete.

Thus, the net utility is

$$D^* = Y_1 - Y_0 - C = [\mu_1(X) - \mu_0(X) - \mu_C(Z)] + (U_1 - U_0 - U_C).$$

- Selection Model

$$D = \mathbf{1}\{D^* \geq 0\}$$

So, in this situation,

$$\text{Cov}(U_1 - U_0 - U_C, D \mid X) \neq 0.$$

A GENERAL MODEL

More general nonlinear outcome model:

$$Y_1 = \mu_1(X) + U_1,$$

$$Y_0 = \mu_0(X) + U_0,$$

$$D^* = \mu_D(Z) - U_D,$$

$$D = \begin{cases} 1, & \text{if } D^* \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- Endogeneity

- Permit that $Cov(U_g, X) \neq 0$, $g \in \{0, 1\}$.
- U_D may not be independent of (U_0, U_1) .
 - Individual resistance to treatment.

- As a normalization, we impose

$$U_D \sim \text{Unif}[0, 1]$$

and

$$\mu_D(Z) = P(Z) \equiv \Pr(D = 1 \mid Z).$$

ASSUMPTIONS

- “Instrumental variable” assumptions

- $Z \mid X$ must provide sufficient variation for identification.
- Z should affect selection decisions, but not the unobserved components of potential outcomes.

(A-1) The term $\mu_D(Z)$ is a nondegenerate random variable conditional on X .

(A-2) The random vectors (U_1, U_D) and (U_0, U_D) are independent of Z conditional on X .

- Technical convenience

(A-3) The distribution of U_D is absolutely continuous with respect to Lebesgue measure.

(A-4) The values of $\mathbb{E}[Y_1]$ and $\mathbb{E}[Y_0]$ are finite.

- Overlap

(A-5)

$$1 > \Pr(D = 1 \mid X) > 0.$$

MOTIVATION

Recall the model we set up:

- U_D represents individual resistance to treatment.
- The model permits

$$\text{Cov}(U_D, U_1 - U_0 \mid X) \neq 0.$$

Therefore, individuals with different levels of resistance to treatment may experience different treatment effects.

Thus, we are interested in

$$\Delta^{MTE}(x, u_D) = \mathbb{E}(Y_1 - Y_0 \mid X = x, U_D = u_D).$$

How should we interpret it?

- The mean gain for individuals with observed characteristics $X = x$ who are indifferent between participating and not participating at u_D .

2. IDENTIFICATION

IDENTIFICATION

$$\begin{aligned}
 \text{Since } E(Y|X, P(Z)=p) &= E(DY_1 + (1-D)Y_0 | X, P(Z)=p) \\
 &= E(Y_0 | X, P(Z)=p) + E(D(Y_1 - Y_0) | X, P(Z)=p) \\
 &= E(Y_0 | X, P(Z)=p) + E(\Delta | X, P(Z)=p, D=1) \cdot (P(D=1 | X, P(Z)=p)) = p \\
 &= E(Y_0 | X, P(Z)=p) + p \cdot E(\Delta | X, P(Z)=p, D=1) \\
 &\stackrel{\text{Conditional independence assumption (A-2)}}{=} E(Y_0 | X) + E(\Delta | X, p \geq U_D) \cdot p
 \end{aligned}$$

换成不可观测的

采用可加性模型

$$\begin{aligned}
 &= E(Y_0 | X) + E(\mu_1(X) - \mu_0(X) + (U_1 - U_0) | X, p \geq U_D) \cdot p \\
 &= E(Y_0 | X) + p[\mu_1(X) - \mu_0(X)] + p \cdot E(U_1 - U_0 | X, p \geq U_D) \quad (*)
 \end{aligned}$$

进一步，如果我们用 MTE 表示 (*) (这种表示更启发式)：

“连续的”

“非连续的”

$$E(Y | X, P(Z)=p) = E(Y_0 | X) + p \Delta^{ATE} + \int_0^p E(U_1 - U_0 | X, U_D = u_D) du_D$$

对 p 求导：

“LIV 估计量”

$$\frac{\partial E(Y | X, P(Z)=p)}{\partial p} = \Delta^{ATE} + E(U_1 - U_0 | U_D = p, X)$$

Thus

$\Delta^{MTE}(p)$

~~$X_1 - X_0$~~

IDENTIFICATION

Step 1: Estimate the propensity score $P(Z) = \Pr(D = 1 \mid Z, X)$.

- Logit or Probit model

Step 2: Estimate $\mathbb{E}(Y \mid X, P) \equiv m(x, p)$.

- Series Estimation

$$Y = X'\beta + \sum_{j=1}^J \gamma_j P^j + \sum_{j=1}^J \Gamma_j X' P^j + \varepsilon$$

- Local estimation

Step 3: Differentiate

$$\widehat{MTE}(x, p) = \frac{\partial \hat{m}(x, p)}{\partial p}.$$

3. RELATIONSHIP WITH OTHER TREATMENT EFFECTS

When $Z \in \{0, 1\}$, recall

$$\Delta^{LATE} \equiv \mathbb{E}(Y_1 - Y_0 \mid D_1 - D_0 = 1).$$

Under the framework of Heckman and Vytlačil (2005), let

$$p_0 = P(Z = 0),$$

$$p_1 = P(Z = 1).$$

Compliers satisfy

$$p_0 < U_D \leq p_1.$$

Therefore,

$$\begin{aligned} \Delta^{LATE}(p_0, p_1) &= \mathbb{E}(Y_1 - Y_0 \mid p_0 < U_D \leq p_1) \\ &= \frac{1}{p_1 - p_0} \int_{p_0}^{p_1} \Delta^{MTE}(u) du. \end{aligned}$$

IV

When Z is continuous, what does linear IV estimate?

Let $J(Z)$ be any instrument satisfying $Cov(J(Z), D | X = x) \neq 0$.
Then the IV estimand is

$$\beta_{IV}(x; J) = \frac{Cov(J(Z), Y | X = x)}{Cov(J(Z), D | X = x)}.$$

It can be shown that

$$\beta_{IV}(x; J) = \int \Delta^{MTE}(x, u_D) \frac{\mathbb{E}(\tilde{J}(Z) | X, P(Z) \geq u_D) \cdot \Pr(P(Z) \geq u_D | X)}{Cov(J(Z), P(Z) | X)} du_D,$$

where

$$\tilde{J}(Z) \equiv J(Z) - \mathbb{E}(J(Z) | X).$$

Define $h_{IV}(u_D | x; J)$. Then

$$\beta_{IV}(x; J) = \int \Delta^{MTE}(x, u_D) h_{IV}(u_D | x; J) du_D.$$

$$\int_0^1 h_{IV}(u_D | x; J) du_D = 1.$$

PROPENSITY SCORE AS IV

When $J(Z) = P(Z)$,

$$h_{IV}(u_D \mid x; P(Z)) = \frac{\mathbb{E}[P(Z) \mid X = x, P(Z) \geq u_D] - \mathbb{E}[P(Z) \mid X = x]}{\text{Var}(P(Z) \mid X = x)} \\ \cdot \Pr(P(Z) \geq u_D \mid X = x).$$

Using the propensity score as an instrument, the weighting function satisfies

$$h_{IV}(u_D \mid x; P(Z)) \geq 0.$$

- $h_{IV} > 0$ if $u_D \in (p_x^{\min}, p_x^{\max})$.
- $h_{IV} = 0$ otherwise.
- $p_x^{\min}$ and $p_x^{\max}$ are respectively the minimum and maximum values in the support of $P(Z) \mid X = x$.

PROPENSITY SCORE AS IV

当 $P(Z)|X=x$ 是连续变量时:

$$\phi_{IV}(X=P(Z)) = \int_{p_{\min}}^{p_{\max}} \Delta^{LIV}(X, u_D) h_{IV}(u_D | X; P(Z)) du_D, \quad \leadsto \text{工具变量 (可被支持) 取决于 } P(Z) \text{ 的支持集 (同 } J(Z))$$

当 $P(Z)|X=x$ 是离散变量时: $p_1 < p_2 < \dots < p_K$

$h_{IV}(u_D | X; P(Z)) \stackrel{= q_j}{\text{is constant}} \text{ in } u_D \text{ over any } (p_j, p_{j+1}) \text{ interval.}$

及令 $\tilde{q}_j = q_j (p_{j+1} - p_j)$,

$$\begin{aligned} \phi_{IV}(X; P(Z)) &= \sum_{j=1}^{K-1} \int_{p_j}^{p_{j+1}} E(\Delta | X=x, U_D=u_D) q_j du_D \\ &= \sum_{j=1}^{K-1} q_j (p_{j+1} - p_j) \int_{p_j}^{p_{j+1}} E(\Delta | X=x, U_D=u_D) \frac{1}{p_{j+1} - p_j} du_D \\ &= \sum_{j=1}^{K-1} \Delta^{LATE}(X, p_j, p_{j+1}) \tilde{q}_j \end{aligned}$$

GENERAL IV

Define

$$T(p \mid x; J) = \mathbb{E}[J \mid P(Z) = p, X = x] - \mathbb{E}[J \mid X = x],$$

and then

$$h_{IV}(u_D \mid x; J) = \frac{\int_{u_D}^1 T(t \mid x; J) dF_{P|X}(t \mid x)}{\text{Cov}(J, P \mid X = x)}.$$

Three properties:

- Two instruments J and J^* generate identical MTE weights at every evaluation point u_D if and only if $\forall X = x, \forall p \in \text{Support}(P(Z) \mid X = x)$,

$$\begin{aligned} & \mathbb{E}[J \mid P(Z) = p, X = x] - \mathbb{E}[J \mid X = x] \\ &= \mathbb{E}[J^* \mid P(Z) = p, X = x] - \mathbb{E}[J^* \mid X = x]. \end{aligned}$$

$\Rightarrow$ Whether two instruments place the same weights on the MTE depends on their relationship with $P(Z)$.

- When is the weight $h_{IV}(u_D \mid x; J)$ nonnegative?

If

$$\mathbb{E}[J \mid P(Z) \geq p, X = x]$$

is weakly monotonic in p , then the weight is nonnegative.

⇒ Whether an instrument assigns positive weights to the MTE depends on its relationship with $P(Z)$.

- The support of $h_{IV}(u_D \mid x; J)$ is contained in $(p_x^{\min}, p_x^{\max})$.

In other words, using the propensity score as an instrument generates nonnegative weights over a broader range of evaluation points than any other instrument.

⇒ The possible support of an instrumental variable estimand is determined by the support of $P(Z)$.

RELATIONSHIP WITH OTHER TREATMENT EFFECTS

TREATMENT EFFECTS AND ESTIMANDS AS WEIGHTED AVERAGES OF THE MARGINAL TREATMENT EFFECT

$$ATE(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) du_D$$

$$TT(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{TT}(x, u_D) du_D$$

$$LATE(x, u_D, u'_D) = \frac{1}{u_D - u'_D} \left[\int_{u'_D}^{u_D} \Delta^{\text{MTE}}(x, u) du \right]$$

$$TUT(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{TUT}(x, u_D) du_D$$

$$PRTE(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{PRTE}(x, u_D) du_D$$

$$IV(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{IV}(x, u_D) du_D$$

$$OLS(x) = \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{OLS}(x, u_D) du_D$$

RELATIONSHIP WITH OTHER TREATMENT EFFECTS

WEIGHTS

$$h_{ATE}(x, u_D) = 1$$

$$h_{TT}(x, u_D) = \left[\int_{u_D}^1 f(p|X=x) dp \right] \frac{1}{E(P|X=x)}$$

$$h_{TUT}(x, u_D) = \left[\int_0^{u_D} f(p|X=x) dp \right] \frac{1}{E((1-P)|X=x)}$$

$$h_{PRTE}(x, u_D) = \left[\frac{F_{P^*,X}(u_D|x) - F_{P,X}(u_D|x)}{\Delta \bar{P}(x)} \right], \text{ where } \Delta \bar{P}(x) = E(P|X=x) - E(P^*|X=x)$$

$$h_{IV}(x, u_D) = \left[\int_{u_D}^1 (p - E(P|X=x)) f(p|X=x) dp \right] \frac{1}{\text{Var}(P|X=x)} \text{ for } P(Z) \text{ as an instrument}$$

$$h_{OLS}(x, u_D) = 1 + \frac{E(U_1|X=x, U_D=u_D)h_1(x, u_D) - E(U_0|X=x, U_D=u_D)h_0(x, u_D)}{\Delta^{\text{MTE}}(x, u_D)},$$

if $\Delta^{\text{MTE}}(x, u_D) \neq 0$,

= 0 otherwise

$$h_1(x, u_D) = \left[\int_{u_D}^1 f(p|X=x) dp \right] \left[\frac{1}{E(P|X=x)} \right]$$

$$h_0(x, u_D) = \left[\int_0^{u_D} f(p|X=x) dp \right] \frac{1}{E((1-P)|X=x)}$$